WQD7003 Data Analytics — Group Project

Predictive Analytics for Mental Health Risk Assessment:

A CRISP-DM Approach

Group 3

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Table of Contents

1.0 Introduction and Business Understanding **1**
1.1 Project Methodology 1
1.2 Project Background 1
1.3 Problem Statement 2
1.4 Project Objectives 2
2.0 Data Understanding **3**
2.1 Dataset Overview 3
2.2 Exploratory Data Analysis 4
2.2.1 Data Quality Assessment 4
2.2.2 Target Variable Analysis 5
2.2.3 Univariate Analysis 6
2.2.4 Bivariate Analysis 8
2.2.5 Multivariate Analysis **10**
3.0 Data Preparation **11**
3.1 Data Cleaning 11
3.2 Feature Engineering 11
3.3 Data Transformation 12
3.3.1 Categorical Variable Encoding **12**
3.3.2 Numerical Variable Standardization 13
3.4 Train-Test Split 13
4.0 Modeling **14**
4.1 Modeling Strategy 14
4.2 Baseline Model Training 14
4.3 Class Balancing Strategy 15
4.4 Hyperparameter Optimization 17
5.0 Evaluation **18**
5.1 Comprehensive Model Performance Comparison 18
5.2 Feature Importance Analysis 18
5.3 ROC Curve Analysis 20
5.4 Confusion Matrix Analysis 21
6.0 Deployment **22**
7.0 Conclusion **23**
8.0 References **24**

1.0 Introduction and Business Understanding

1.1 Project Methodology

This project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework to conduct research. CRISP-DM is a systematic and iterative data science project methodology comprising six main phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. This structured approach ensures research rigor and reproducibility, making it particularly suitable for public health applications. The cyclical nature of the CRISP-DM framework allows us to continuously optimize and adjust strategies based on findings as the project progresses, ensuring that the final model truly meets practical application needs.

1.2 Project Background

Mental health has emerged as one of the most critical public health challenges of the 21st century. According to World Health Organization statistics, approximately 1.1 billion people globally were living with mental disorders in 2025 [1], with anxiety and depressive disorders being the two most common types. Despite continuous advances in treatment research, early detection of mental health disorders remains a fundamental barrier to effective intervention. Traditional screening processes primarily rely on self-reporting and clinical interviews, which often only become effective after symptoms have already manifested and struggle to comprehensively consider the complex interactions among various factors. Hence, there is an urgent need for scalable, data-driven approaches to identify at-risk individuals earlier, enabling timely intervention and prevention.

1.3 Problem Statement

Current mental health screening systems have significant limitations. First, traditional methods heavily rely on self-report questionnaires, which are susceptible to social desirability bias where individuals underreport symptoms or exaggerate positive attributes[2]. Second, these screenings typically occur only after symptoms become apparent, missing optimal windows for early intervention [3]. Third, conventional approaches struggle to systematically integrate and analyze complex interactions among demographic, lifestyle, and health-related factors [3]. These limitations result in many high-risk individuals not being identified timely, delaying treatment opportunities and increasing disease burden. Therefore, developing an automated predictive system using machine learning that comprehensively considers multidimensional factors to identify high-risk populations has important practical significance and social value.

1.4 Project Objectives

This project aligns with United Nations Sustainable Development Goal 3 (Good Health and Well-being) and aims to leverage machine learning techniques to develop a predictive solution for early mental health risk identification. The project establishes three core objectives: First, to develop high-accuracy binary classification models capable of accurately predicting whether individuals have a history of mental illness based on demographic, lifestyle, and health-related factors. Second, to identify and quantify key risk factors most closely associated with mental illness history, providing empirical evidence for clinical decision-making and public health policy formulation.

Third, to determine the most suitable modeling approach for mental health risk prediction through comparison of multiple machine learning algorithms performance.

2.0 Data Understanding

2.1 Dataset Overview

The dataset used in this study comes from Kaggles "Mental Health Depression Dataset," a large-scale comprehensive dataset specifically designed for mental health research. The dataset contains 413,768 observational samples covering 15 predictive variables and 1 target variable, providing a solid data foundation for in-depth analysis of mental health risk factors.

Feature	Description
Sample Size	413,768
Number of Predictive Variables	15
Target Variable	History of Mental Illness (Yes/No)
Numerical Variables	3 (Age, Income, Number of Children)
Categorical Variables	12
Class Distribution	Approximately 70% No mental illness history, 30% Yes

Table 2.1: Data Description

The dataset has a reasonable distribution of feature types, with numerical variables including Age, Income, and Number of Children. Categorical variables cover Employment Status, Education Level, Marital Status, Physical Activity Level, Dietary

Habits, Sleep Patterns, Smoking Status, Alcohol Consumption, History of Substance Abuse, Family History of Depression, and Chronic Medical Conditions.

Name	Age	Marital Status	Education Level	Number of Children	Smoking Status	Physical Activity Level	Employment Status	Income	Alcohol Consumption	Dietary Habits	Sleep Patterns	History of Mental Illness	History of Substance Abuse	Family History of Depression	Chronic Medical Conditions
Christine Barker	31	Married	Bachelor's Degree	2	Non-smoker	Active	Unemployed	26265.67	Moderate	Moderate	Fair	Yes	No	Yes	Yes
Jacqueline Lewis	55	Married	High School	1	Non-smoker	Sedentary	Employed	42710.36	High	Unhealthy	Fair	Yes	No	No	Yes
Shannon Church	78	Widowed	Master's Degree	1	Non-smoker	Sedentary	Employed	125332.79	Low	Unhealthy	Good	No	No	Yes	No
Charles Jordan	58	Divorced	Master's Degree	3	Non-smoker	Moderate	Unemployed	9992.78	Moderate	Moderate	Poor	No	No	No	No
Michael Rich	18	Single	High School	0	Non-smoker	Sedentary	Unemployed	8595.08	Low	Moderate	Fair	Yes	No	Yes	Yes

Figure 2.1: First few examples of the dataset

2.2 Exploratory Data Analysis

2.2.1 Data Quality Assessment

Before beginning in-depth analysis, we first conduct a comprehensive data quality assessment. The data quality check results are satisfactory, there are no missing values in the dataset, with all 413,768 records being complete. Additionally, no duplicate records were found.

Check Item	Result
Total Missing Values	0
Number of Duplicate Records	0
Dataset Shape	(413768, 16)

Table 2.2: Data Quality Check Results

Descriptive statistics of numerical variables reveal the basic characteristic distribution of the dataset. From the statistical summary, we can see that the age variable ranges from 18 to 80 years, with a mean age of 49 years and standard deviation of 18.16

years, showing that the sample covers all age groups of the adult population. The income variable shows considerable variability, ranging from a minimum of $0.41 to a maximum of $21,001.03, alongside high standard deviation. The income variable contains outliers, but these outliers represent genuinely existing high-income individuals and are therefore retained in the analysis. The number of children distribution is relatively concentrated, ranging from 0 to 4, with a mean of 1.3 and median of 1.

Statistic	Age	Number of Children	Income
count	413768.00	413768.00	413768.00
mean	49.00	1.30	50661.71
std	18.16	1.24	40624.10
min	18.00	0.00	0.41
25%	33.00	0.00	21001.03
50%	49.00	1.00	37520.13
75%	65.00	2.00	76616.30
max	80.00	4.00	209995.22

Table 2.3:Statistical Summary

2.2.2 Target Variable Analysis

Among the 413,768 samples, 287,943 samples (approximately 69.6%) reported no history of mental illness, while 125,825 samples (approximately 30.4%) reported having a history of mental illness.

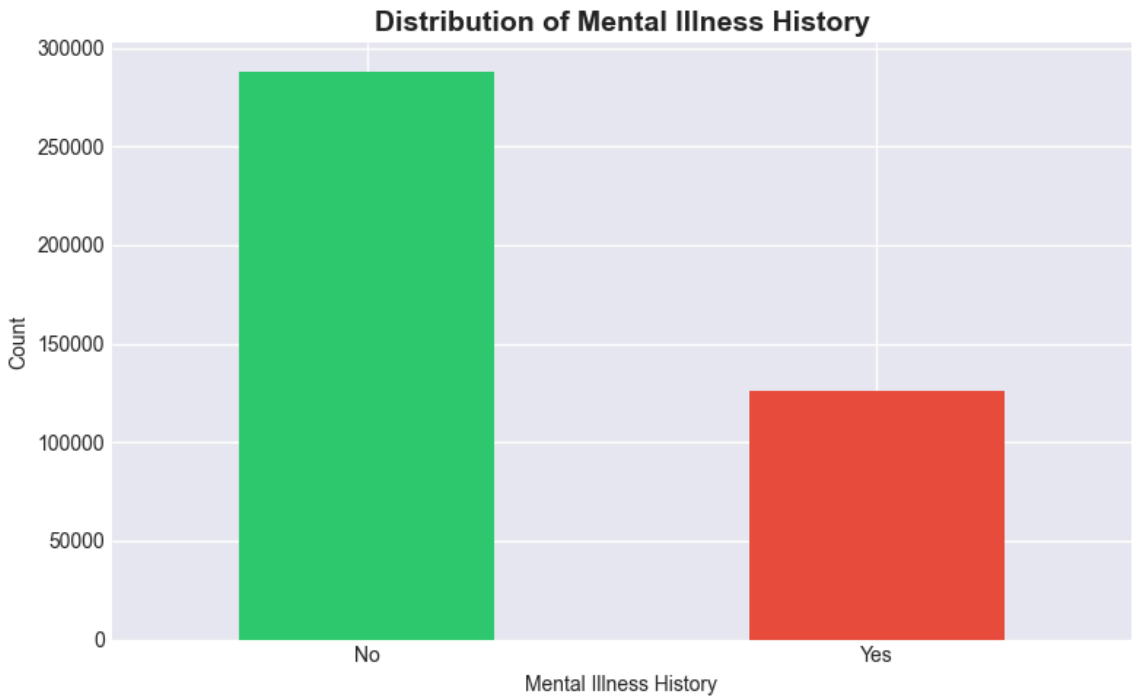

Figure 2.2: Distribution of Target Variable

This class imbalance distribution is common in mental health research and reflects the actual prevalence of mental illness in the general population. However, this imbalance also means that certain strategies need to be adopted during model training to avoid the model being biased toward the majority class.

2.2.3 Univariate Analysis

Distribution analysis of numerical variables reveals the demographic characteristics of the sample. The age distribution is relatively uniform, covering all age groups from young adults to elderly. The income distribution shows right skewness, with most samples income concentrated in lower value ranges, which aligns with real-world income distribution characteristics. The number of children distribution shows that most samples

have no children or only a few children, with sample numbers gradually decreasing as the number of children increases.

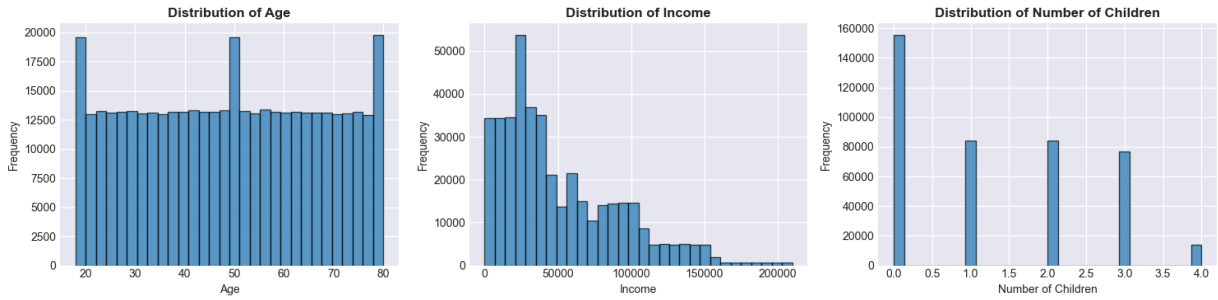

Figure 2.3: Distribution of Numerical Variables

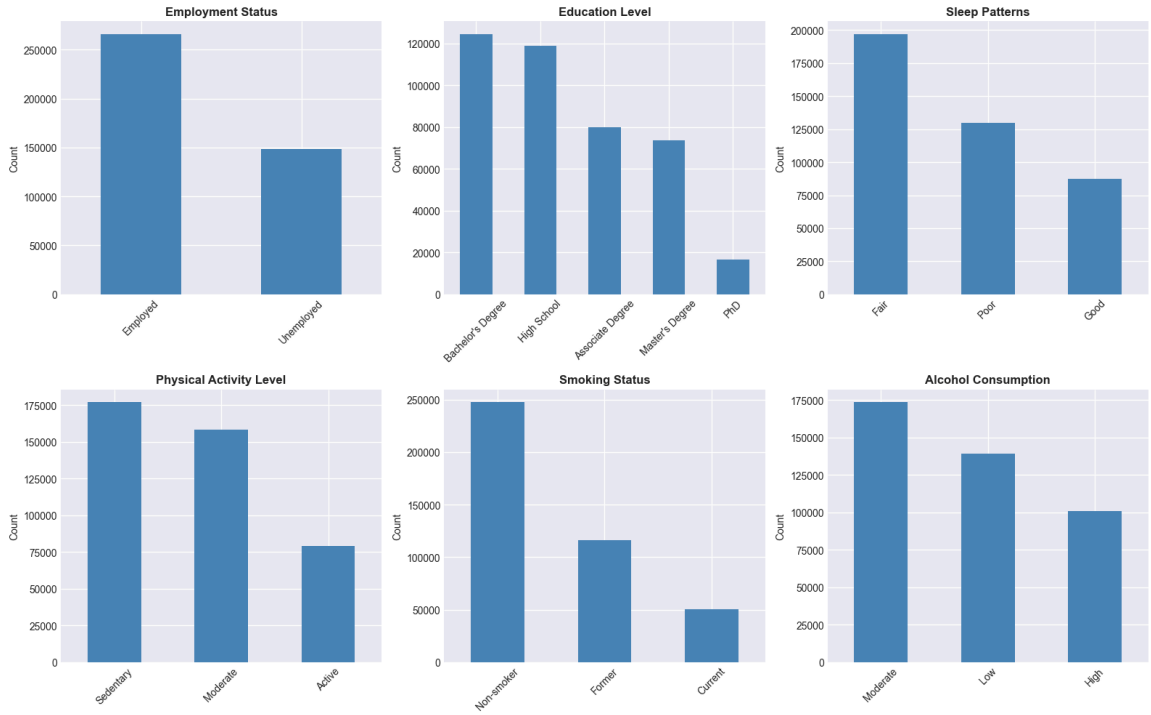

Figure 2.4: Distribution of Categorical Variables

Categorical variable distribution analysis shows that all categorical variables do not exhibit uniform distribution but rather display their own characteristic patterns. Variables such as employment status, education level, sleep patterns, physical activity level,

smoking status, and alcohol consumption all show reasonable distributions, and the diversity of these variables provides possibilities for identifying associations between lifestyle and mental health.

2.2.4 Bivariate Analysis

Bivariate analysis focuses on exploring relationships between each predictive variable and the target variable. Through boxplot and t-test analysis of the relationship between numerical variables and the target variable, we found significant statistical associations.

Variable	t-statistic	p-value	Significance
Age	15.8818	8.81×10^{-57}	Highly Significant
Income	-88.8344	0.0000	Highly Significant
Number of Children	1.7453	0.0809	Not Significant

Table 2.4: T-test Results

From the t-test results, we can see that age and income have highly significant statistical associations with mental illness history, while the association with number of children is not significant.

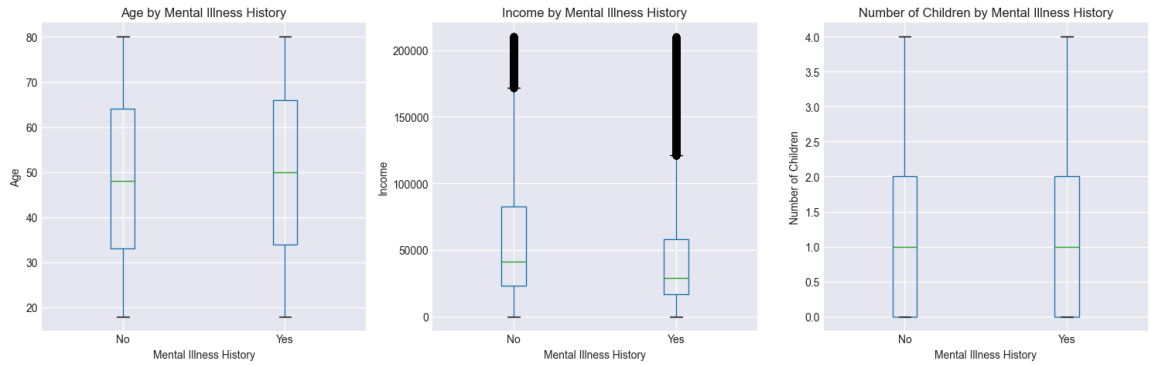

Figure 2.5: Boxplots of Numerical Variable against Target Variable

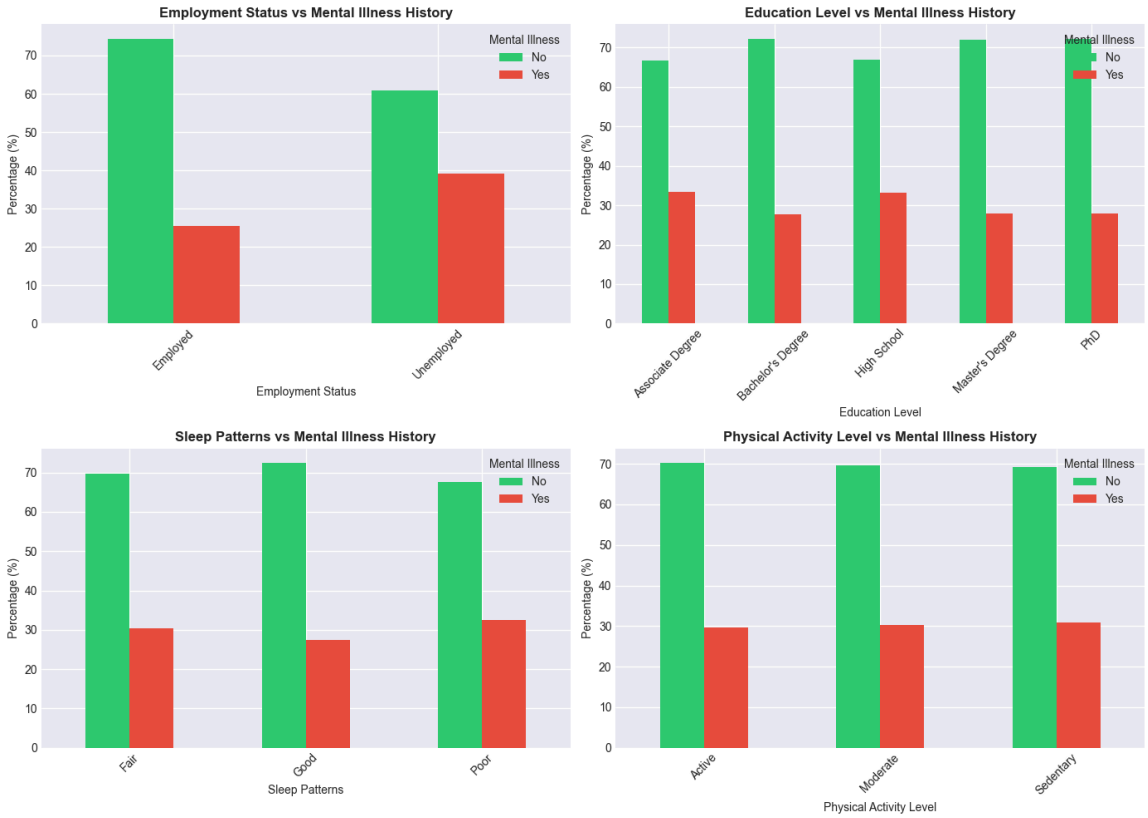

Figure 2.6: Relationship between selected Categorical Variable and Target Variable

Employment status indicates strong association between unemployment and mental illness, while lower education levels are associated with higher mental illness risk,

and poor sleep quality is closely related to higher mental illness prevalence. Finally, lower physical activity levels are associated with mental illness.

2.2.5 Multivariate Analysis

Correlation analysis among numerical variables shows that the three numerical variables (age, income, number of children) exhibit weak correlations, indicating that these variables are relatively independent without serious multicollinearity issues.

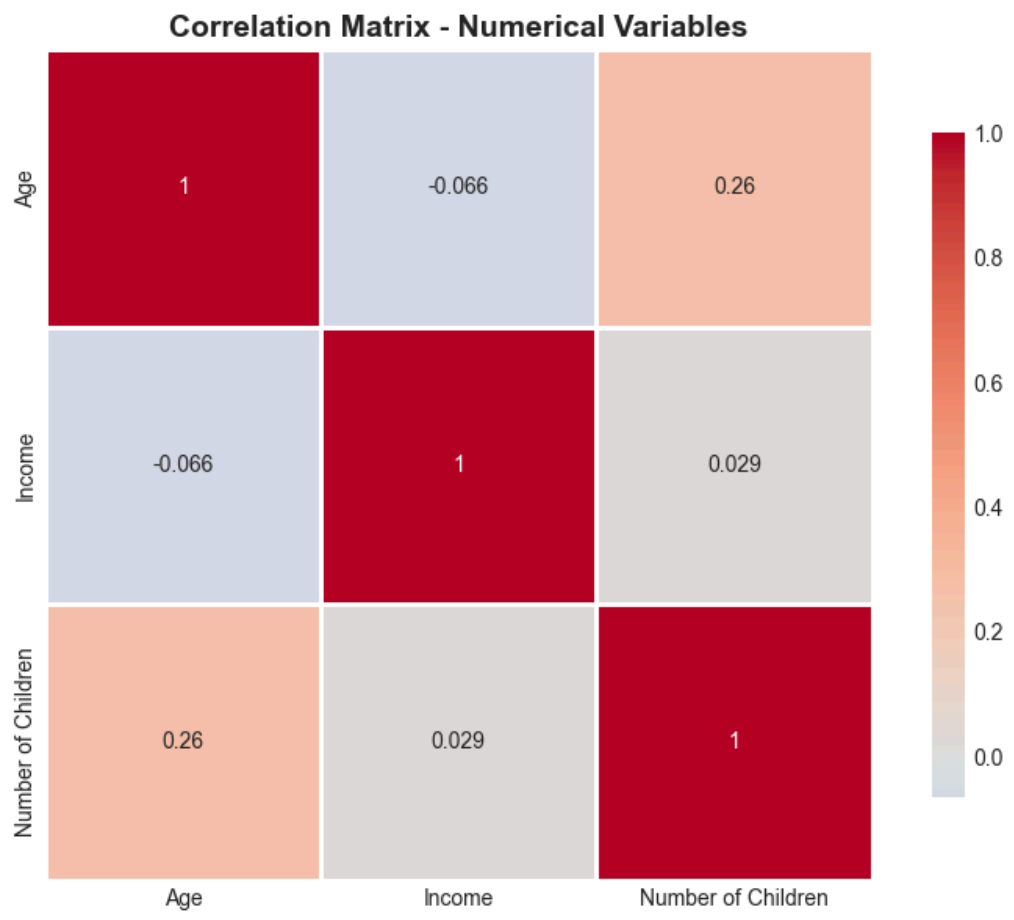

Figure 2.7: Relationship between selected Categorical Variable and Target Variable

3.0 Data Preparation

3.1 Data Cleaning

Based on findings from exploratory data analysis, minimal cleaning work was required due to high data quality. For outliers in the income variable, we conducted a detailed assessment. Using the Interquartile Range (IQR) method, we identified 5,157 outliers (accounting for 1.25% of total samples). After careful consideration, we decided to retain these outliers as they represent genuinely existing high-income individuals, and removing them might lead to information loss and affect the models generalization ability.

Item	Value
Number of Income Outliers	5,157
Outlier Proportion	1.25%
Processing Decision	Retain (represent genuine high-income individuals)

Table 3.1: Outlier Assessment Results

3.2 Feature Engineering

Feature engineering is one of the core strategies for improving model performance. Based on domain knowledge and insights from exploratory data analysis, we created three new derived features to enrich the models learning capabilities.

Created Features	Approach
Age Group	Discretized continuous age variable into four categories (Young Adult, Adult, Middle Age and Senior)
Income Category	Converted continuous income variable into three levels (Low, Medium, High)
Risk Factor Count	Count of multiple known risk factors per individual

Table 3.2: Created Features

After feature engineering completion, the dataset shape expanded from (413768, 16) to (413768, 19), adding 3 features. These new features not only enable the model to capture mental health characteristics across different life stages and socioeconomic levels but also capture the cumulative effect of risk factors through risk factor counting.

3.3 Data Transformation

3.3.1 Categorical Variable Encoding

Categorical variable encoding is a necessary step to convert non-numerical data into formats processable by machine learning algorithms. First, we separated features and the target variable. The target variable "History of Mental Illness" was converted to binary numerical values (Yes=1, No=0) using label encoding.

For predictive variables, we identified 14 categorical features and 4 numerical features. Categorical features include Name, Marital Status, Education Level, Smoking Status, Physical Activity Level, etc. Numerical features include Age, Number of Children, Income, and Risk_Factor_Count. We implemented label encoding for

categorical features, a nominal encoding method that converts each category to a unique integer.

3.3.2 Numerical Variable Standardization

Numerical variable standardization is crucial for the performance of many machine learning algorithms. We used StandardScaler to standardize all numerical features, transforming each feature to a distribution with mean 0 and standard deviation 1. This standardization ensures that features with different scales have equal weight in model training.

	mean-original	std-original	mean-scaled	std-scaled
Age	49.00	18.16	0.0	1.0
Number of Children	1.30	1.24	-0.0	1.0
Income	50661.71	40624.10	0.0	1.0
Risk_Factor_Count	2.31	0.46	0.0	1.0

Figure 3.1: Numerical variables before and after standardisation

From the comparison results, we can see that standardization successfully transformed all numerical features to distributions with mean 0 and standard deviation 1, demonstrating the effectiveness of using the standard z-score method.

3.4 Train-Test Split

To objectively evaluate model performance and avoid overfitting, we split the dataset into training and test sets. Using an 80:20 split ratio and stratified sampling strategy, we ensured that the class distribution of the target variable in both training and test sets remained consistent with the original dataset.

The class distribution in training and test sets remains consistent with the original dataset, ensuring fairness in model evaluation. A total of 18 features are used for modeling, all in numerical format, including encoded categorical variables and standardized numerical variables.

4.0 Modeling

4.1 Modeling Strategy

This study adopts a model comparison strategy, selecting five machine learning algorithms with different characteristics for mental health risk prediction. The five selected algorithms are: Logistic Regression, Random Forest, XGBoost, LightGBM and CatBoost.

Training Divided into Three Phases:

- Phase 1: Train baseline models using default configurations
- Phase 2: Introduce class balancing strategy to address data imbalance
- Phase 3: Hyperparameter tuning for the best model

4.2 Baseline Model Training

The first phase of the modeling process is to establish baseline models, training all five algorithms with default parameters to obtain initial performance benchmarks. All models use the same training data and random seed (random_state=42) to ensure reproducibility.

Model	Recall	AUC-ROC
Logistic Regression	0.0000	0.5920
Random Forest	0.0695	0.5659
XGBoost	0.0111	0.5916
LightGBM	0.0003	0.5968
CatBoost	0.0054	0.5938

Table 4.1 Baseline Model Evaluation Metrics

Baseline model training results show that all models AUC-ROC scores range between 0.56 and 0.59, indicating that models have certain predictive ability but relatively limited performance. These results provide benchmark references for subsequent model optimization.

4.3 Class Balancing Strategy

To address the class imbalance problem (70% no mental illness history vs 30% with mental illness history), we introduced a class weight adjustment strategy in the second phase. By setting the class_weight=balanced parameter, the algorithm automatically adjusts sample weights based on the inverse of class frequency, giving minority class samples higher weight in the loss function.

Model	Recall	AUC-ROC
Logistic Regression	0.4788	0.5920
Random Forest	0.0567	0.5671

Model	Recall	AUC-ROC
XGBoost	0.5923	0.5921
LightGBM	0.5383	0.5968
CatBoost	0.5745	0.5941

Table 4.2 Class-Balanced Model Evaluation Metrics

The introduction of class balancing strategy significantly improved recall, which is a key metric for identifying high-risk individuals (minority class). Gradient boosting algorithms (XGBoost, LightGBM, CatBoost) showed the best performance after class balancing, with recall reaching 0.5383 and AUC-ROC improving to 0.5968. Logistic Regressions recall was 0.4788, while Random Forest's recall was only 0.0567, showing poor performance.

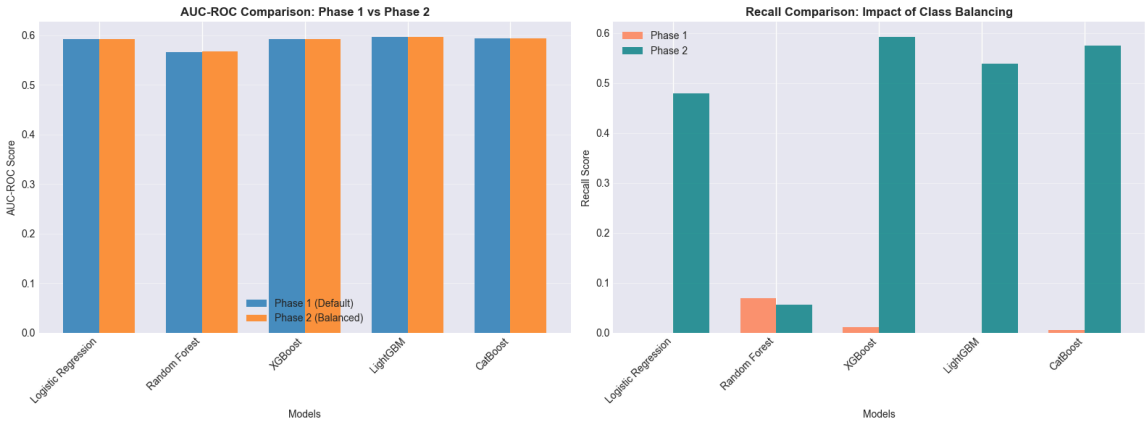

Figure 4.1 Performance of models before and after class-balanced

4.4 Hyperparameter Optimization

Based on Phase 2 results, LightGBM was identified as the best-performing model (AUC-ROC: 0.5968). Therefore, we conducted hyperparameter optimization for LightGBM to further improve performance.

Parameter	Search Range
n_estimators	[100, 200]
max_depth	[5, 10, 15]
learning_rate	[0.01, 0.1]
num_leaves	[31, 50]

Table 4.3 Hyperparameter Search Space

We used Grid Search combined with 3-fold cross-validation to find the optimal parameter combination. The search space contained 24 candidate parameter combinations, totaling 72 model fits (24 combinations × 3 folds). The model trained with optimal parameters performed as follows on training and test sets:

Dataset	AUC-ROC
Training Set	0.6069
Test Set	0.5969

Table 4.4 Performance of LightGBM after optimisation of hyperparameters

The difference between training set AUC (0.6069) and test set AUC (0.5969) is only 0.01, indicating that the model has good generalization ability without overfitting.

5.0 Evaluation

5.1 Comprehensive Model Performance Comparison

Through three phases of systematic training and optimization, we conducted a comprehensive evaluation of all models' performance. The comparison between Phase 1 (baseline models) and Phase 2 (class balancing) reveals the importance of class balancing strategy, as there was a significant improvement in terms of recall. In terms of AUC-ROC, after introducing the class weight balancing method, most models AUC-ROC scores slightly decreased or remained stable.

5.2 Feature Importance Analysis

To understand which factors contribute most to mental health risk prediction, we used SHAP (SHapley Additive exPlanations) value analysis to quantify each features contribution. SHAP values are a game theory-based model interpretation method that can provide feature-level explanations for each prediction. The SHAP summary plot shows the direction and magnitude of each features impact on model predictions.

According to SHAP value analysis, Unemployment status emerged as the most influential predictor in the model, followed by Income and Smoking Status. This indicates that that economic factors play a core role in mental health. Unemployed individuals typically have no income or very low income, and this economic pressure is associated with various mental health problems, including financial stress, food

insecurity, housing instability, etc. which all of which may lead to or exacerbate mental health issues.

Figure 5.1: SHAP value analysis

5.3 ROC Curve Analysis

The ROC (Receiver Operating Characteristic) curve is an important tool for evaluating discriminative abilities.

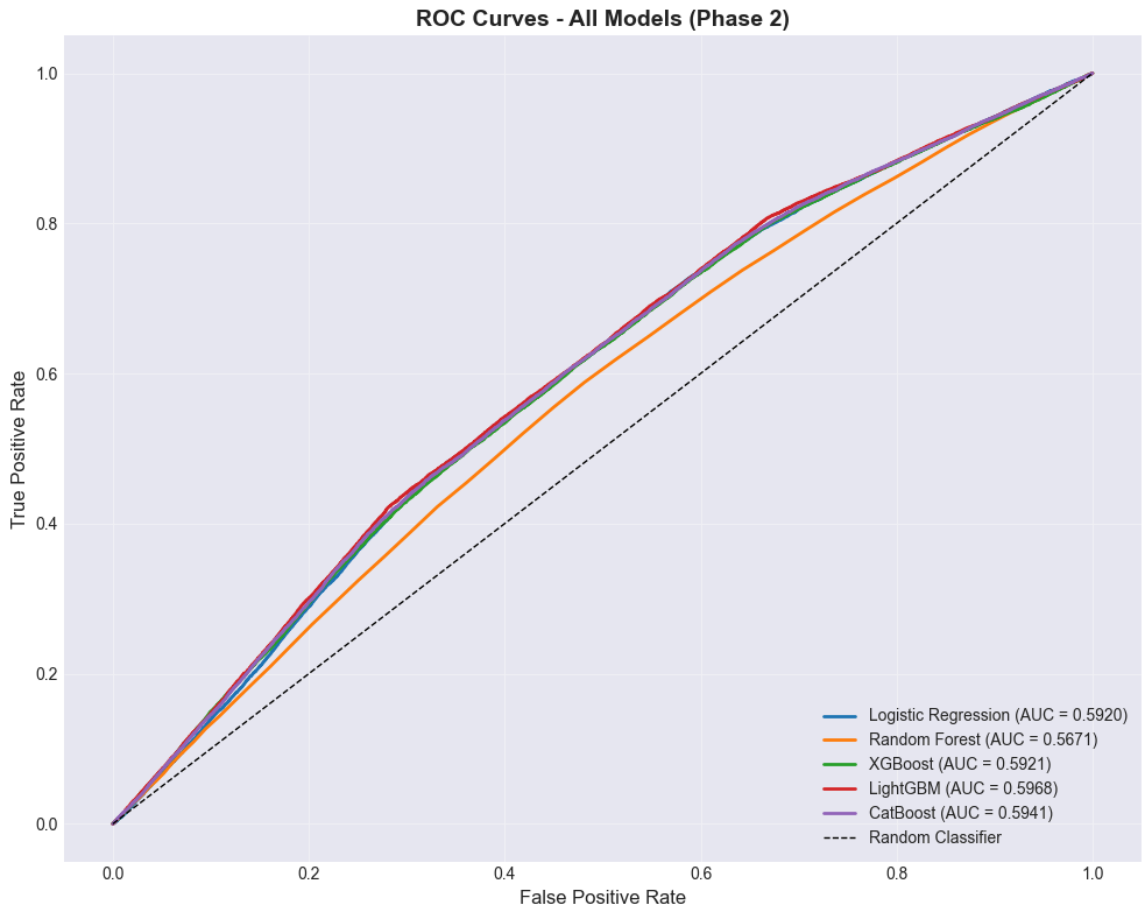

Figure 5.2: AUC-ROC Scores for All Models

All models show similar discriminative ability (AUC approximately 0.59-0.60), with gradient boosting methods (LightGBM, CatBoost, XGBoost) slightly outperforming traditional approaches. These curves allow us to identify optimal classification thresholds to balance true positive rate and false positive rate. Considering the complexity of mental

health prediction and data limitations, these results are reasonable, and the models do perform better than random guessing (AUC=0.5).

5.4 Confusion Matrix Analysis

The confusion matrix provides a detailed breakdown of model prediction results, showing the specific numbers of true positives, true negatives, false positives, and false negatives.

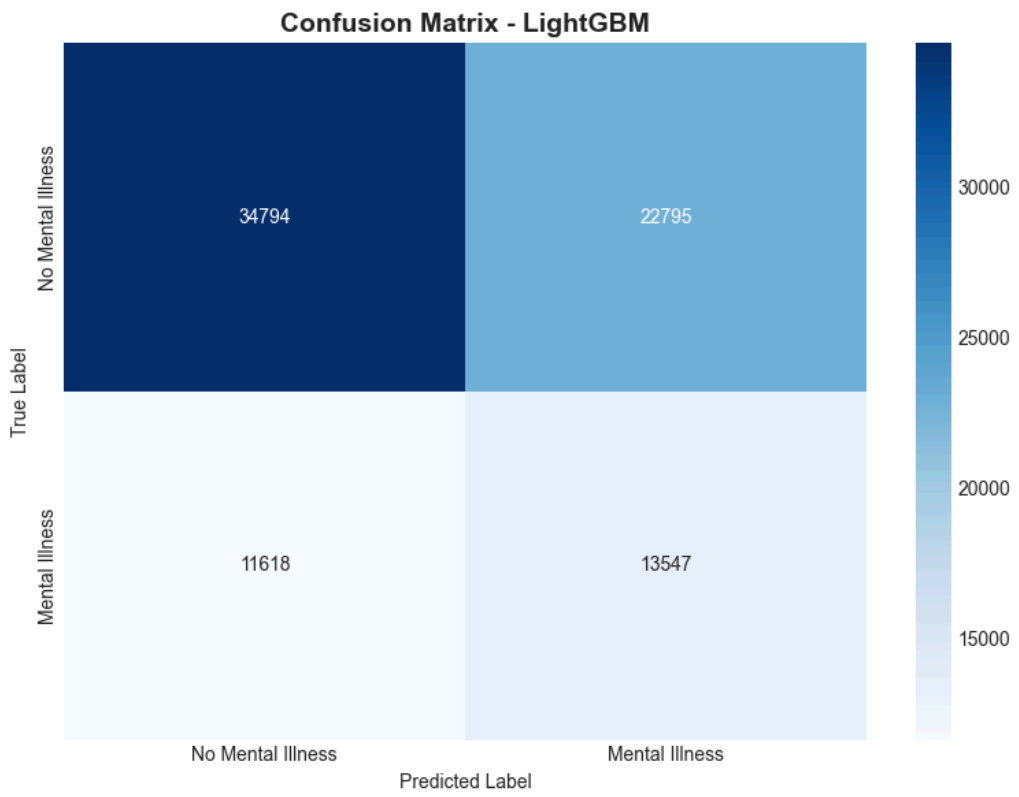

Figure 5.3: Confusion Matrix for LightGBM (Phase 2)

Overall Performance: The model correctly classified 48,341 cases out of 82,754 total predictions, representing approximately 58.4% accuracy. The model's recall (0.5383) and specificity (0.6042) are relatively balanced, which is a direct result of the

class weight balancing strategy. Without class balancing, the model would tend to predict the majority class, resulting in extremely low recall.

6.0 Deployment

The mental health risk prediction model developed in this study aims to provide practical risk screening tools for related stakeholders. Deployment methods can adopt API service format for integration into existing health information systems, or be developed as standalone web applications for use by medical professionals and the public.

To ensure the model maintains good performance in practical applications, a comprehensive monitoring plan needs to be established. It is recommended to track key performance metrics (such as AUC-ROC and recall) monthly, monitor changes in feature distributions to identify data drift, and retrain the model quarterly using new data. Model performance should be maintained above a certain threshold of AUC-ROC ensure prediction quality.

When deploying mental health prediction models, ethics and privacy protection are primary considerations. Health data privacy compliance must be ensured, adhering to relevant data protection regulations. To mitigate algorithmic bias, regular demographic fairness audits of the model are needed to ensure balanced performance across different populations. The model should provide interpretable prediction results, such as showing prediction basis to clinicians through SHAP values, enhancing model transparency and credibility. Most importantly, model outputs should serve as support tools for clinical

judgment rather than replacements, always maintaining the "Human-in-the-Loop" principle, ensuring final decisions are made by professionals.

7.0 Conclusion

This study successfully applied the CRISP-DM methodology to develop a machine learning-based mental health risk prediction model, providing a data-driven solution for early identification of high-risk individuals for mental health issues, aligning with United Nations Sustainable Development Goal 3 (Good Health and Well-being). The study used a dataset containing 413,768 samples. In the data preparation phase, we cleaned, engineered, and transformed all records, creating 18 modeling features, including new derived features.

By evaluating five machine learning algorithms (Logistic Regression, Random Forest, XGBoost, LightGBM, and CatBoost) and employing a three-phase training strategy (baseline models, class balancing, hyperparameter optimization), we identified LightGBM as the best model. The introduction of class weight balancing strategy significantly improved recall from near zero to 0.5383. The final optimized model achieved an AUC-ROC of 0.5969 on the test set, with only a 0.01 difference between training and test sets, demonstrating good generalization ability. Through SHAP analysis, we revealed that unemployment status, income, and smoking status are the most important predictive factors.

As for the limitations, firstly, the dataset comes from a specific platform and may have selection bias, requiring further validation of sample representativeness. Second, the cross-sectional study design can only identify associations rather than establish causal

relationships. Third, the target variable "mental illness history" is a broad category that does not distinguish specific types of mental disorders.

With respect to future direction, richer clinical features such as clinical assessments, treatment history, medication history, and genetic factors can be introduced to improve predictive ability. Next, explore alternative imbalanced data handling techniques (such as SMOTE, ADASYN) and deep learning methods (such as neural networks), as well as ensemble stacking techniques to improve model performance. Through these continuous improvements and validations, mental health risk prediction models will be better able to serve public health practice and contribute to achieving global mental health goals.

8.0 References

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